

CNF

Question -1) Remove unit productions from the Cfg below

- a) $W' \rightarrow W$
 $W \rightarrow TWY \mid t \mid TW \mid WY$
 $T \rightarrow tTW \mid t \mid tW$
 $Y \rightarrow WyW \mid T \mid yy$
- b) $S' \rightarrow S$
 $S \rightarrow RSR \mid rT \mid RS \mid SR \mid r$
 $R \rightarrow T \mid S$
 $T \rightarrow t$
- c) $S \rightarrow A \mid B$
 $A \rightarrow aAs \mid a \mid bb \mid AS$
 $B \rightarrow A \mid aa \mid SbS$
 $C \rightarrow c$

Question -2) Eliminate null productions from the Cfg below

- a) $W' \rightarrow W$
 $W \rightarrow TWY \mid t \mid TW \mid WY$
 $T \rightarrow tTW \mid t \mid \epsilon$
 $Y \rightarrow WyW \mid T \mid yy$
- b) $S \rightarrow A \mid B$
 $A \rightarrow aAs \mid a \mid bb \mid \epsilon$
 $B \rightarrow A \mid aa \mid SBS \mid \epsilon$
 $C \rightarrow c$
- c) $S' \rightarrow S$
 $S \rightarrow RSR \mid rT \mid RS \mid SR \mid \epsilon$
 $R \rightarrow T$
 $T \rightarrow t$

Question -3) Eliminate useless variables from the Cfg below

- a) $S \rightarrow aAa \mid bBb \mid aa$
 $A \rightarrow a$
 $B \rightarrow b$
 $C \rightarrow a \mid b \mid ab$
- b) $W' \rightarrow a$
 $W \rightarrow TW \mid t \mid TW$
 $T \rightarrow tTW \mid t \mid \epsilon$
 $Y \rightarrow c$

- c) $S \rightarrow A \mid B$
 $E \rightarrow A \mid B$
 $A \rightarrow aAs \mid a \mid bb \mid \epsilon$
 $B \rightarrow A \mid aa \mid SBS \mid \epsilon$
 $C \rightarrow \epsilon$
 $D \rightarrow a \mid b$

Question -4) Convert the following CFG to CNF (Chomsky Normal form).

a) Start symbol S

$$\begin{aligned} S &\rightarrow aAa \mid bBb \mid \epsilon \\ A &\rightarrow C \mid a \\ B &\rightarrow C \mid b \\ C &\rightarrow CDE \mid \epsilon \\ D &\rightarrow A \mid B \mid ab \end{aligned}$$

b) Start symbol S

$$\begin{aligned} S &\rightarrow AB \mid CA \\ A &\rightarrow a \\ B &\rightarrow BC \mid AB \\ C &\rightarrow aB \mid b \end{aligned}$$

c) Start symbol S

$$\begin{aligned} S &\rightarrow ASB \mid \epsilon \\ A &\rightarrow aAS \mid a \\ B &\rightarrow SbS \mid A \mid bb \end{aligned}$$

d) Start symbol S

$$\begin{aligned} S &\rightarrow 0A0 \mid 1B1 \mid BB \\ A &\rightarrow C \\ B &\rightarrow S \mid A \\ C &\rightarrow S \mid \epsilon \end{aligned}$$

e) Start symbol S

$$\begin{aligned}
 S &\rightarrow AAA \mid B \\
 A &\rightarrow aA \mid B \\
 B &\rightarrow \epsilon
 \end{aligned}$$

f) Start symbol S

$$\begin{aligned}
 S &\rightarrow AB \mid CD \\
 A &\rightarrow BC \mid a \\
 B &\rightarrow AC \mid C \\
 C &\rightarrow AB \mid CD \\
 D &\rightarrow AC \mid d
 \end{aligned}$$

g) $S \rightarrow aSb \mid b$

CYK

Question -5)

Grammar	Strings to check
$S \rightarrow AB \mid BC$ $A \rightarrow BA \mid a$ $B \rightarrow CC \mid b$ $C \rightarrow AB \mid a$	a) w = ababb b) w = baaba c) w = ababa d) w = baaab e) w = aabab f) w = ababaa

Question -6)

$S_0 \rightarrow WB \mid b$

S WB |b

A a

B b

W AS

String to check: **aabbbb**